

AMPV V2 Supplementary Numerical Detail

This supplement preserves the detailed descriptive values and trend listings removed from the compact main-text candidate. All content is copied or reformatted from the frozen S51 manuscript state; no new simulations or scientific values are introduced.

0.1 Table S1. Complete cell-resolved numerical summaries

0.1.1 $C = 0$

	K_ref	Workflow	Median	e	Vulnerability	
1	A		0.048	10.3%	0.0061	7.024
1	B		0.059	12.0%	0.0050	7.024
1	C		0.255	36.0%	-0.0015	7.024
1	D		0.060	12.3%	-0.0011	7.024
1	E		0.012	4.3%	8.972e-04	7.024
2	A		0.024	6.3%	0.0031	14.047
2	B		0.029	6.3%	0.0025	14.047
2	C		0.125	20.7%	0.0015	14.047
2	D		0.030	8.0%	-0.0014	14.047
2	E		0.0060	1.7%	-1.792e-04	14.047
5	A		0.0096	2.7%	0.0012	35.119
5	B		0.012	3.0%	0.0010	35.119
5	C		0.049	10.0%	-0.0031	35.119
5	D		0.013	3.3%	-0.0017	35.119
5	E		0.0024	0.3%	-2.627e-04	35.119
10	A		0.0048	0.7%	6.141e-04	70.237
10	B		0.0058	1.0%	5.025e-04	70.237
10	C		0.026	7.3%	-0.0033	70.237
10	D		0.0064	1.0%	-0.0017	70.237
10	E		0.0012	0.0%	-1.551e-04	70.237
20	A		0.0024	0.0%	3.071e-04	140.474
20	B		0.0029	0.0%	2.513e-04	140.474
20	C		0.012	3.3%	-0.0041	140.474
20	D		0.0038	0.0%	-0.0016	140.474
20	E		6.177e-04	0.0%	-7.780e-05	140.474

0.1.2 $C = 0.6$

	K_ref	Workflow	Median	e	Vulnerability	
1	A		0.072	13.3%	0.065	7.024
1	B		0.062	13.7%	0.038	7.024
1	C		0.265	34.3%	0.016	7.024
1	D		0.057	12.3%	0.037	7.024
1	E		0.014	5.7%	0.0070	7.024
2	A		0.037	6.0%	0.033	14.047

	K_ref	Workflow	Median	e	Vulnerability	
2	B		0.031	6.0%	0.019	14.047
2	C		0.130	20.7%	0.010	14.047
2	D		0.029	7.3%	0.018	14.047
2	E		0.0066	3.0%	0.0032	14.047
5	A		0.015	3.0%	0.013	35.119
5	B		0.013	3.3%	0.0078	35.119
5	C		0.052	11.0%	0.0012	35.119
5	D		0.011	3.0%	0.0056	35.119
5	E		0.0027	0.7%	0.0012	35.119
10	A		0.0074	1.3%	0.0067	70.237
10	B		0.0063	1.0%	0.0039	70.237
10	C		0.027	6.0%	-7.745e-05	70.237
10	D		0.0053	0.7%	0.0017	70.237
10	E		0.0013	0.0%	5.867e-04	70.237
20	A		0.0037	0.3%	0.0033	140.474
20	B		0.0032	0.3%	0.0020	140.474
20	C		0.012	4.3%	-7.970e-04	140.474
20	D		0.0032	0.3%	-2.582e-04	140.474
20	E	6.540e-04	0.0%	2.940e-04		140.474

0.1.3 C = 1

	K_ref	Workflow	Median	e	Vulnerability	
1	A		0.118	19.3%	0.114	7.024
1	B		0.070	14.3%	0.059	7.024
1	C		0.286	36.0%	0.047	7.024
1	D		0.067	14.0%	0.055	7.024
1	E		0.015	6.3%	0.011	7.024
2	A		0.060	11.7%	0.059	14.047
2	B		0.036	7.3%	0.030	14.047
2	C		0.137	19.3%	0.026	14.047
2	D		0.034	8.0%	0.027	14.047
2	E		0.0075	3.7%	0.0054	14.047
5	A		0.025	4.7%	0.024	35.119
5	B		0.014	3.0%	0.012	35.119
5	C		0.057	9.3%	0.010	35.119
5	D		0.013	3.3%	0.0094	35.119
5	E		0.0030	1.0%	0.0021	35.119
10	A		0.012	2.3%	0.012	70.237
10	B		0.0072	1.0%	0.0061	70.237
10	C		0.027	7.3%	0.0036	70.237
10	D		0.0061	1.0%	0.0035	70.237
10	E		0.0015	0.0%	9.953e-04	70.237
20	A		0.0062	0.3%	0.0060	140.474
20	B		0.0036	0.3%	0.0030	140.474

	K_ref	Workflow	Median	e	Vulnerability
20	C		0.014	4.0%	4.754e-04
20	D		0.0028	0.3%	7.874e-04
20	E		7.387e-04	0.0%	5.012e-04

0.2 Supplementary Note S1. Reference-contrast trend listing

Across the frozen K_ref grid, the descriptive trends are:

- C=0, workflow A: median |e| is nonincreasing across K_ref (0.048 at K_ref=1; 0.0024 at K_ref=20); vulnerability is nonincreasing (10.3% -> 0.0%).
- C=0, workflow B: median |e| is nonincreasing across K_ref (0.059 at K_ref=1; 0.0029 at K_ref=20); vulnerability is nonincreasing (12.0% -> 0.0%).
- C=0, workflow C: median |e| is nonincreasing across K_ref (0.255 at K_ref=1; 0.012 at K_ref=20); vulnerability is nonincreasing (36.0% -> 3.3%).
- C=0, workflow D: median |e| is nonincreasing across K_ref (0.060 at K_ref=1; 0.0038 at K_ref=20); vulnerability is nonincreasing (12.3% -> 0.0%).
- C=0, workflow E: median |e| is nonincreasing across K_ref (0.012 at K_ref=1; 6.177e-04 at K_ref=20); vulnerability is nonincreasing (4.3% -> 0.0%).
- C=0.6, workflow A: median |e| is nonincreasing across K_ref (0.072 at K_ref=1; 0.0037 at K_ref=20); vulnerability is nonincreasing (13.3% -> 0.3%).
- C=0.6, workflow B: median |e| is nonincreasing across K_ref (0.062 at K_ref=1; 0.0032 at K_ref=20); vulnerability is nonincreasing (13.7% -> 0.3%).
- C=0.6, workflow C: median |e| is nonincreasing across K_ref (0.265 at K_ref=1; 0.012 at K_ref=20); vulnerability is nonincreasing (34.3% -> 4.3%).
- C=0.6, workflow D: median |e| is nonincreasing across K_ref (0.057 at K_ref=1; 0.0032 at K_ref=20); vulnerability is nonincreasing (12.3% -> 0.3%).
- C=0.6, workflow E: median |e| is nonincreasing across K_ref (0.014 at K_ref=1; 6.540e-04 at K_ref=20); vulnerability is nonincreasing (5.7% -> 0.0%).
- C=1, workflow A: median |e| is nonincreasing across K_ref (0.118 at K_ref=1; 0.0062 at K_ref=20); vulnerability is nonincreasing (19.3% -> 0.3%).
- C=1, workflow B: median |e| is nonincreasing across K_ref (0.070 at K_ref=1; 0.0036 at K_ref=20); vulnerability is nonincreasing (14.3% -> 0.3%).
- C=1, workflow C: median |e| is nonincreasing across K_ref (0.286 at K_ref=1; 0.014 at K_ref=20); vulnerability is nonincreasing (36.0% -> 4.0%).
- C=1, workflow D: median |e| is nonincreasing across K_ref (0.067 at K_ref=1; 0.0028 at K_ref=20); vulnerability is nonincreasing (14.0% -> 0.3%).
- C=1, workflow E: median |e| is nonincreasing across K_ref (0.015 at K_ref=1; 7.387e-04 at K_ref=20); vulnerability is nonincreasing (6.3% -> 0.0%).

0.3 Supplementary Note S2. Contamination trend listing

At fixed reference contrast, changing C modifies the observed error and vulnerability patterns. The cell-resolved contamination trends are:

- K_ref=1, workflow A: median |e| is nondecreasing with C (0.048 at C=0; 0.118 at C=1); vulnerability is nondecreasing (10.3% -> 19.3%).

- $K_{ref}=1$, workflow B: median $|e|$ is nondecreasing with C (0.059 at $C=0$; 0.070 at $C=1$); vulnerability is nondecreasing (12.0% $\rightarrow$ 14.3%).
- $K_{ref}=1$, workflow C: median $|e|$ is nondecreasing with C (0.255 at $C=0$; 0.286 at $C=1$); vulnerability is nonmonotonic (36.0% $\rightarrow$ 36.0%).
- $K_{ref}=1$, workflow D: median $|e|$ is nonmonotonic with C (0.060 at $C=0$; 0.067 at $C=1$); vulnerability is nondecreasing (12.3% $\rightarrow$ 14.0%).
- $K_{ref}=1$, workflow E: median $|e|$ is nondecreasing with C (0.012 at $C=0$; 0.015 at $C=1$); vulnerability is nondecreasing (4.3% $\rightarrow$ 6.3%).
- $K_{ref}=2$, workflow A: median $|e|$ is nondecreasing with C (0.024 at $C=0$; 0.060 at $C=1$); vulnerability is nonmonotonic (6.3% $\rightarrow$ 11.7%).
- $K_{ref}=2$, workflow B: median $|e|$ is nondecreasing with C (0.029 at $C=0$; 0.036 at $C=1$); vulnerability is nonmonotonic (6.3% $\rightarrow$ 7.3%).
- $K_{ref}=2$, workflow C: median $|e|$ is nondecreasing with C (0.125 at $C=0$; 0.137 at $C=1$); vulnerability is nonincreasing (20.7% $\rightarrow$ 19.3%).
- $K_{ref}=2$, workflow D: median $|e|$ is nonmonotonic with C (0.030 at $C=0$; 0.034 at $C=1$); vulnerability is nonmonotonic (8.0% $\rightarrow$ 8.0%).
- $K_{ref}=2$, workflow E: median $|e|$ is nondecreasing with C (0.0060 at $C=0$; 0.0075 at $C=1$); vulnerability is nondecreasing (1.7% $\rightarrow$ 3.7%).
- $K_{ref}=5$, workflow A: median $|e|$ is nondecreasing with C (0.0096 at $C=0$; 0.025 at $C=1$); vulnerability is nondecreasing (2.7% $\rightarrow$ 4.7%).
- $K_{ref}=5$, workflow B: median $|e|$ is nondecreasing with C (0.012 at $C=0$; 0.014 at $C=1$); vulnerability is nonmonotonic (3.0% $\rightarrow$ 3.0%).
- $K_{ref}=5$, workflow C: median $|e|$ is nondecreasing with C (0.049 at $C=0$; 0.057 at $C=1$); vulnerability is nonmonotonic (10.0% $\rightarrow$ 9.3%).
- $K_{ref}=5$, workflow D: median $|e|$ is nonmonotonic with C (0.013 at $C=0$; 0.013 at $C=1$); vulnerability is nonmonotonic (3.3% $\rightarrow$ 3.3%).
- $K_{ref}=5$, workflow E: median $|e|$ is nondecreasing with C (0.0024 at $C=0$; 0.0030 at $C=1$); vulnerability is nondecreasing (0.3% $\rightarrow$ 1.0%).
- $K_{ref}=10$, workflow A: median $|e|$ is nondecreasing with C (0.0048 at $C=0$; 0.012 at $C=1$); vulnerability is nondecreasing (0.7% $\rightarrow$ 2.3%).
- $K_{ref}=10$, workflow B: median $|e|$ is nondecreasing with C (0.0058 at $C=0$; 0.0072 at $C=1$); vulnerability is constant (1.0% $\rightarrow$ 1.0%).
- $K_{ref}=10$, workflow C: median $|e|$ is nondecreasing with C (0.026 at $C=0$; 0.027 at $C=1$); vulnerability is nonmonotonic (7.3% $\rightarrow$ 7.3%).
- $K_{ref}=10$, workflow D: median $|e|$ is nonmonotonic with C (0.0064 at $C=0$; 0.0061 at $C=1$); vulnerability is nonmonotonic (1.0% $\rightarrow$ 1.0%).
- $K_{ref}=10$, workflow E: median $|e|$ is nondecreasing with C (0.0012 at $C=0$; 0.0015 at $C=1$); vulnerability is constant (0.0% $\rightarrow$ 0.0%).
- $K_{ref}=20$, workflow A: median $|e|$ is nondecreasing with C (0.0024 at $C=0$; 0.0062 at $C=1$); vulnerability is nondecreasing (0.0% $\rightarrow$ 0.3%).
- $K_{ref}=20$, workflow B: median $|e|$ is nondecreasing with C (0.0029 at $C=0$; 0.0036 at $C=1$); vulnerability is nondecreasing (0.0% $\rightarrow$ 0.3%).
- $K_{ref}=20$, workflow C: median $|e|$ is nonmonotonic with C (0.012 at $C=0$; 0.014 at $C=1$); vulnerability is nonmonotonic (3.3% $\rightarrow$ 4.0%).
- $K_{ref}=20$, workflow D: median $|e|$ is nonincreasing with C (0.0038 at $C=0$; 0.0028 at $C=1$); vulnerability is nondecreasing (0.0% $\rightarrow$ 0.3%).
- $K_{ref}=20$, workflow E: median $|e|$ is nondecreasing with C ($6.177e-04$ at $C=0$; $7.387e-04$ at $C=1$); vulnerability is constant (0.0% $\rightarrow$ 0.0%).

0.4 Table S2. Lowest tested K_ref with vulnerability $\leq 5\%$

Workflow	C=0	C=0.6	C=1
A	5	5	5
B	5	5	5
C	20	20	20
D	5	5	5
E	1	2	2

The threshold summary is descriptive and valid only on the tested grid.